

5. Developments in plastic technology have allowed the production of bioplastics using plants as the source of raw material.

The table below compares the production of the same mass of a bioplastic with polystyrene. Polystyrene is made from crude oil.

	Bioplastic	Polystyrene
Energy consumption (kWh)	10.7	22.6
Water used in production (dm ³)	74.6	185
Waste for landfill produced (g)	1.74	2.10
Biodegradable	Yes	No
Breakdown products	carbon dioxide, water and methane	styrene
kgCO ₂ eq/kg produced	0.50	2.2

Use your knowledge and the information in the table to answer the following questions.

- (a) Explain **two** advantages to the environment of using bioplastics. [4]

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- (b) Explain **one** environmental disadvantage of using bioplastics. [2]

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- (c) Manufacturers of this bioplastic claim that it is a carbon-neutral product. State the additional information that is needed to support this claim. [2]

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